

Endless Garden

Fall 2025 ME Executive Summary

Design Team

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Abstract

Approximately 54 million Americans face food insecurity and another 23.5 million live in areas where fresh produce is unavailable or unaffordable. Indoor hydroponics can help address this problem, but current products can be costly, may require past hydroponic knowledge, and opt for batch harvesting where everything is planted and harvested at once. This can create alternating periods of excess production and no production, which does not allow consumers to have consistent fresh produce. This paper presents the Endless Garden, an automated hydroponic system that will, once a week, have ready to eat produce through staggered planting. This allows those who lack time and technical knowledge to have access to constant freshly grown produce. Our prototype demonstrates the growth of a relatively small crop, little gem romaine, allowing the design team to test and iterate within a manageable scope. All parts of the design are scalable, so any kind of crop can be accommodated with relatively simple changes. The paper details the design, testing, and integration of four subsystems, the plug refill system which automatically moves a new seedling into position when a mature plant is harvested, a nutrient delivery system with pumps and sensors that keeps the water at the right pH (5.8-6.0), a water level controller that maintains proper root submersion, and an LCD interface that informs users when plants are ready to harvest or when nutrients or water needs to be added. We developed this system through iterative prototyping, first designing and testing each subsystem individually before integrating them into the complete assembly. Testing confirms successful automated plug replacement and reliable nutrient delivery, demonstrating the system's potential to provide hands-off gardening experience. This work establishes a foundation for accessible food production to anyone regardless of gardening or hydroponic knowledge.

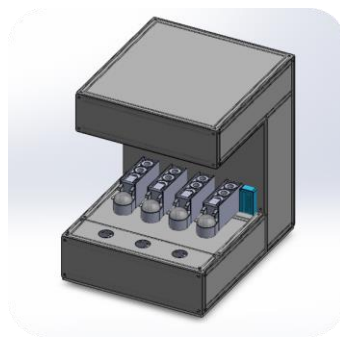


Figure 1: Isometric View of Prototype CAD

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Need

Around 54 million people in the United States are food insecure as of 2022, and many people face barriers to accessing healthy food such as fresh produce. Approximately 23.5 million Americans live in areas that either do not have a grocery store nearby or the price of the food is too high [1]. However, growing healthy produce is not a realistic option for busy people living in small apartments; they lack sufficient space, time, or resources. Indoor gardening has the potential to overcome these barriers and provide significant benefits in all types of homes. The perceived potential market of indoor gardening is evident in the wide variety of existing products created to provide fresh food in limited or unconventional spaces. Existing designs offer features such as hydroponic systems, smart watering, and pre-seeded soil pods [2-6]. However, none of these products has achieved widespread adoption because many consumers lack gardening knowledge, have nearby grocery stores, and find the high-cost risks of gardening intimidating [7]. Our product, the Endless Garden, will bring a new perspective on these problems by further exploring personal automated gardening. The goal of this project is to let people grow lettuce at home using a semi-automated system that can function with little user interaction. In the future, this technology can be further developed to be fully automated.

Background and Significant Prior Work

Introduction

This section provides insight into an understanding of hydroponic systems, existing market solutions, and patents currently being used in the automated agriculture space. The research done for Endless Garden helped the approach to the development of water tanks, the pod growth systems as well as user interaction with the product. This resulted in informed design of the pod refill, plant monitoring, nutrient, and water tank subassemblies.

Deep Water Culture

The Endless Garden design is modeled on one of the simplest and most used hydroponics methods, deep water culture (DWC). It differs from soil-based growth by replacing soil with a large tank of oxygenated water and nutrient solution. These deep-water hydroponic systems simplify plumbing compared to other aeroponic misting systems by having plants sit in net-pots that are suspended in water 6-inch-deep water solution mixed with a nutrient [9]. The water these plants sit in is continuously oxygenated with an air stone to prevent hypoxia, a condition where roots die from lack of oxygen. Hydroponic systems give greater access for the roots to take in nutrients, helping accelerate growth by 50% compared to conventional soil-based agriculture [8]. As many hydroponic systems are used indoors, they replace the need for natural sunlight with artificial light. All plants have an ideal range of light that they need; for example, baby gem romaine needs a continuous stream of light over a period of 12-14 hours a day at a distance of at least 6 inches from the leaves to ensure ideal growth rates and prevent burning the plant [8]. Maintaining water chemistry is also critical for plant development, and baby gem requires a pH in the range of 5.8-6.0 pH to allow proper absorption of nutrients from the water [8]. Many current solutions require the user to control the pH of the water manually.

Current Solutions

A wide range of indoor gardening systems currently exist, each offering unique features aimed at making plant care more accessible, automated, and efficient [1-6]. LetPot is an indoor gardening system that also approaches automated gardening via pre-packaged pods ready to grow. This product features a hydroponic solution, meaning that the plant roots are submerged in water instead of soil, making for a mess-free

experience. Whenever the water level for the roots gets too low, the user gets alerted through the LetPot mobile app [3]. The base models of LetPot range from about \$70–\$200.

Gardyn is another standing, large-scale indoor gardening product that introduces a hybrid growing approach combining hydroponic and aeroponic principles. The system supports the growth of up to 30 large plants simultaneously and uses AI-driven sensors and cameras to monitor plant health and environmental conditions. With dedicated zones for root development and leaf growth, Gardyn offers “high-density vertical growth” making it suitable in indoor urban environments. It includes a five-gallon water tank that supports longer-term maintenance, providing a scalable and largely autonomous indoor gardening experience. In contrast to brands like LetPot, Gardyn does not currently offer an “affordable” base model. The starting price is \$500 with more advanced models surpassing \$1,000 [4].

All models mentioned follow a “batch” process for harvesting, meaning users plant all produce at once and wait for a full growth cycle before harvesting. After harvesting, the entire system resets and is replanted [1-6]. Another similarity across these brands is the use of pre-packaged starter pods, which contain seeds, a growing medium, and nutrients tailored for each system. This approach simplifies setup by eliminating the need to purchase separate soil, fertilizer, or other gardening supplies. Each of these systems aims to bridge areas of automation, yield, aesthetics, and usability—and together they showcase the diversity and innovation currently available in the indoor gardening market.

Patents

To understand the technological developments in the hydroponics landscape, it is important to look at existing patents. US 20240138328, introduced a compact, frame-based indoor garden featuring removable trays, each with multiple plant containers fed by a shared liquid reservoir. Users could slide individual trays in and out for harvesting or maintenance without disrupting the rest of the system [10]. US 7055282 B2, described a vertically stacked hydroponic column composed of interlocking modules. Each module contained a hollow interior with a perforated base that channeled nutrient solution through a central conduit [11]. US 20180007845 A1, covered an aeroponic system where plants were supported in sealed root chambers within a housing. Stalk-conforming seals isolated the canopy, while embedded sensors monitored root-zone conditions. Air-assisted nozzles sprayed atomized nutrient solution directly onto the roots, and an integrated control system regulated temperature and moisture to maintain precise growth parameters [12].

Summary

The background research helped inform Endless Gardens of critical system design as well as its functionality. Deep water culture hydroponics provides the foundation helping accelerate growth by working around the essential parameters of the Baby Gem Romain, with 6-inch-deep water, continuous oxygenation, 12-14 hours of light and precise PH control. Analyzing the current market helped inform current solutions to design constraints, with LetPot and Gardyn that offered only a limited automation that struggles by offering a batched style harvesting method that creates either excess at one moment in time or large production gaps. They also have systems with a lot of manual control, particularly around its PH control, limiting the market to more experienced users. The patents help reveal established methods for efficient nutrient and water delivery; with a focus on scalable architecture and automated environmental management. By understanding the current innovations and limitations, we can pinpoint areas such as continuous harvest cycles and integrated mold control features where Endless Garden could introduce novel solutions and differentiated capabilities.

Design Solution

Introduction

Our Design Solution is the Endless Garden, shown in Figure 1. This countertop device comprises an environmentally- and nutritionally controlled hydroponic tank, a sensor-based refill system that allows seed-soil plugs to be automatically loaded into the tank, a low maintenance water supply, grow lighting, and a graphic user interface. Briefly, the user loads twelve germination pods into the device and fills the supply tank with 3.2 gallons of water. One pod is immediately loaded into the tank, and others follow at staggered intervals. At the end of an initial growth cycle of 7.5 weeks, fresh produce is produced once per week. Specific subsystems, mechanisms, and electronics are described in this section.

Plug Refill - Mechanical

When a mature plant is ready to be harvested, the user will be notified to manually remove the plug, which holds the nutrient sponge and the plant. After removal, a new, pre-seeded plug will need to take its place to restart the growing cycle. This refill process is designed to be as automated as possible to allow for a “hands-off” experience for the user. Our design for this system is gravity fed meaning the pods are all sitting at a angle and are pushed forward by gravity. With this mechanism as our inspiration, we decided we will have 1 plug growing and 3 plugs in reserve, supported by a rail system shown in Figure 2. This allows a total of 4 growing cycles per plug refill station, ensuring that when a plug is removed, it can be automatically replaced 3 times before a restock is needed.

The plug that holds the actively growing plant will sit in the extruded circle shown in Figure 2 above. To limit interference between or the growing plugs during the plant growth cycle, there will be a door separating the stored plugs from the growing plugs. This door will stop the forward force of the stored plugs. The door design mimics that of a Pez dispenser and can open and close to allow or block a pod from moving into the growth pod area. The door will open using a small servo motor once the user removes a pod. To allow germination to occur, a germination lid will also be included in the system and be controlled by another servo motor. These processes are detailed in the electrical component overview below. In our prototype, there will be four of these systems, allowing 4 little gem Romaines to grow at a time while 12 are stored.



Figure 2: Plug Refill 3D Print

Plug Refill – Electrical

Once the user harvests a fully grown head of lettuce, a reflective infrared (IR) sensor (ITR20001/T) detects that the pod has been removed. The IR sensor determines whether a pod is in place by emitting infrared light onto the top surface of the pod and measuring how much of that light returns to its phototransistor. When a pod is in front of the sensor, the reflected light keeps the phototransistor conducting current to ground, and the sensor reports a steady reading that the code interprets as “pod present”. Once the user removes the pod, the reflected light level drops, the phototransistor stops conducting, and the sensor output changes to “pod removed”. When the sensor detects that a pod has been removed, it waits three seconds, and then an SG90 servo motor rotates the door holding the unused pods in place. Then, after a new pod is pushed into place, the door will rotate back to its original position. The code executes the pod-replacement sequence only when it detects this specific transition from ‘pod present’ to ‘pod removed’, ensuring that the system responds once per removal instead of repeatedly. We added the 3-second delays

to the code to allow the user time to remove the pod and give the pod an appropriate amount of time to fall into position. Figure 3 outlines this logic.

The rotating doors and germination lids use SG90 servo motors, which have a starting current of 1 A. Our system contains four pod-refill stations, and each station uses two servo motors (one motor controls the door, and another controls the germination lid), leading to a total of eight motors in this system. Although the system staggers the growth cycles, and it is not likely that more than two motors will start up at the same time, several motors could still start up simultaneously if multiple stations need to be refilled at the same time. The worst-case scenario would be if all eight motors tried to start at the same time and demanded up to 8 A, which exceeds the 4 A limit of our power supply. However, to lower the overall current draw from the motors and ensure the power supply will not fail, each motor is connected to an NPN transistor such that the motor's ground is only connected to the microcontroller unit's (MCU) ground when the transistor closes. This prevents the motors from drawing current unless the MCU explicitly activates them. In the appendix, Figure 18 shows the schematics for the IR sensors and motors, and Figure 19 shows the wiring diagram for the motors and transistors.

In addition to the electronics that automate pod delivery, the design includes a sensor system that alerts the user if the pod refill station needs restocking. Our system uses a Hall effect sensor that reads the distance of a neodymium magnet attached to a “pusher” behind the four waiting pods. As the pod supply decreases, the magnet moves closer to the sensor. Once the

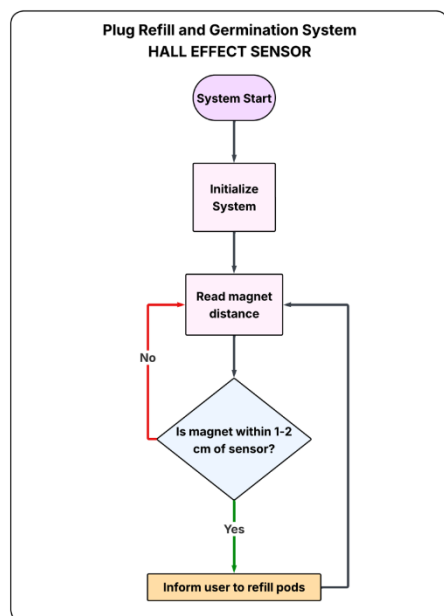


Figure 4: Hall Effect Sensor Logic

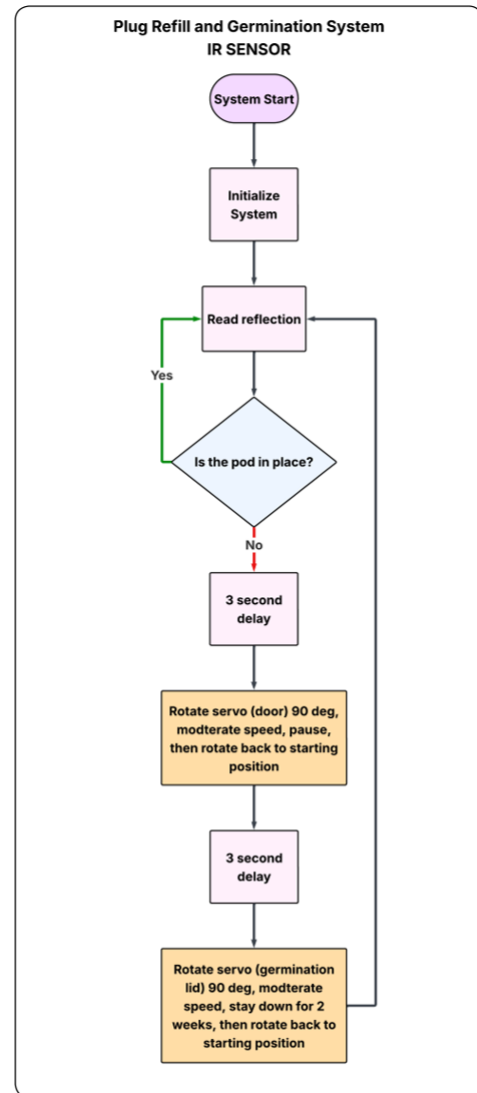


Figure 3: IR Sensor Logic

magnet on the door comes within 1-2 cm of the sensor, indicating that the supply of unused pods is nearly empty, it will send a signal to the user informing them that the station needs to be refilled. Figure 4 outlines this logic.

Nutrient Controls – Function

As stated in the Background section, the circulation of supplementary nutrients is required in hydroponic systems. Since the pH of tap water is variable, acid solutions known as pH down and basic solutions known as pH up are used to correct for the deviation. The target pH for our system is 5.8-6.0.

Nutrient Controls – Electrical

The nutrient solution, pH down solution, and pH up solution are held in separate glass jars and controllably moved into the front

tank through peristaltic pumps. The pH sensor communicates the proper information to the relays and consequently switches the pumps on and off. The pumps are supplied 12V through a power supply which is stepped down to 5V through a buck converter. The relay schematic to turn the pumps on and off is shown in Figure 10 and the wiring diagram is shown in Figure 11, located in the “Design Process and Results” section.

Nutrient Controls – Mechanical

Three glass jars and silicone tubing are used to route the liquid into the front tank for plant absorption. These materials were selected for their noncorrosive properties within the operating pH range. The jars and pumps are mounted in place by custom 3D printed fixtures as shown in Figure 5 below.

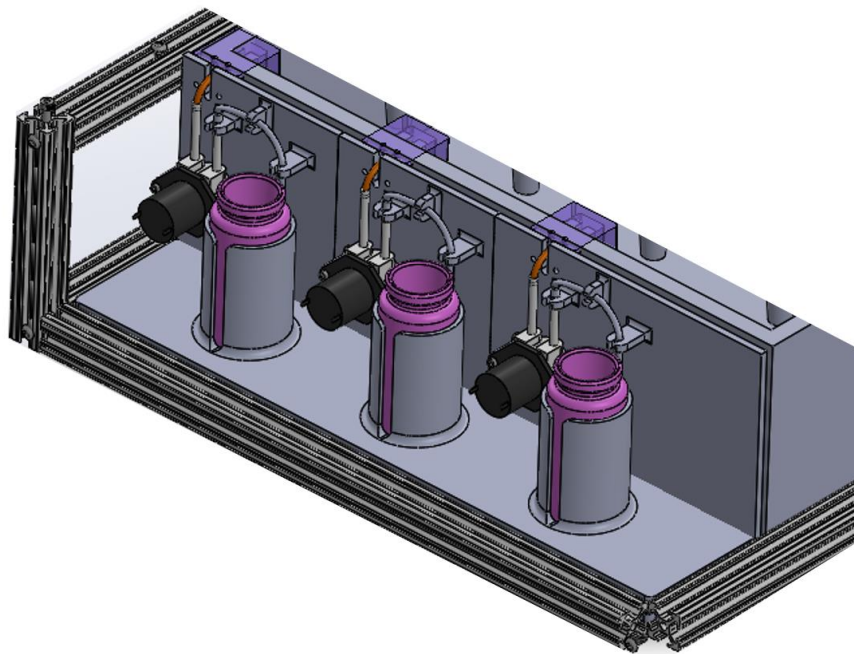


Figure 5: Nutrient Pump and Container Fixtures

Water Level Controls – Function

The water level control subsystem was designed to maintain a consistent water height in the growing reservoir, ensuring optimal root submersion and oxygen exposure. In our system, the plants sit in a Deep Water Culture tank where the roots must remain submerged in approximately 6 inches of nutrient solution for healthy growth. To maintain this height, the water level should roughly sit right below the top rim of the reservoir lid, where the net pots are mounted. This spacing ensures that the base of each pod is just above the waterline while the lower roots remain submerged, preventing over-saturation or root rot.

Water Level Controls – Mechanical

Our team implemented a dual-reservoir system controlled by a normally closed solenoid valve. The top reservoir serves as the backup supply, and the bottom reservoir is the active grow tank. When the water level in the bottom tank drops due to evaporation or plant uptake, a float sensor or ultrasonic sensor triggers the solenoid valve to open. This allows water to flow from the upper reservoir, through a check valve and tubing, into the lower reservoir by gravity until the target level is reached. Once the sensor detects that the water has returned to its set height, the solenoid closes to prevent overfilling or backflow.

The decision to use a normally closed valve ensures safety. If power is lost or the control system fails, the valve remains shut, avoiding uncontrolled flooding. This method also simplifies plumbing since the top tank provides natural pressure from gravity, eliminating the need for an electric pump between reservoirs. The check valve and anti-siphon loop were added to prevent unintended draining or reverse flow when the solenoid closes.

Plant Monitoring Controls

The team considered many ideas when monitoring plant growth. Informing users when their plants reached maturity through automated means rather than visual inspection proved paramount to the design. Users unfamiliar with growth patterns of baby gem romaine could end up harvesting their produce prematurely. To solve this problem, the team integrated an IR break beam sensor that conveys harvest readiness information automatically.

To implement the design the electrical component was created with the IR break beam sensor operating as a two-part system with an emitter bulb and a receiver. Using the configuration, the team designed the system so that once the plant reached 6 inches above the seed, this would inform the user that the plant was ready to harvest. A Tredix IR sensor was chosen thanks to its 1m maximum effective range and low profile.

Framing

We will use 1"x1" 8020 extrusions to mount all our components. Sheet metal and acrylic will then be mounted to all exterior sides as shown in Figure 6.

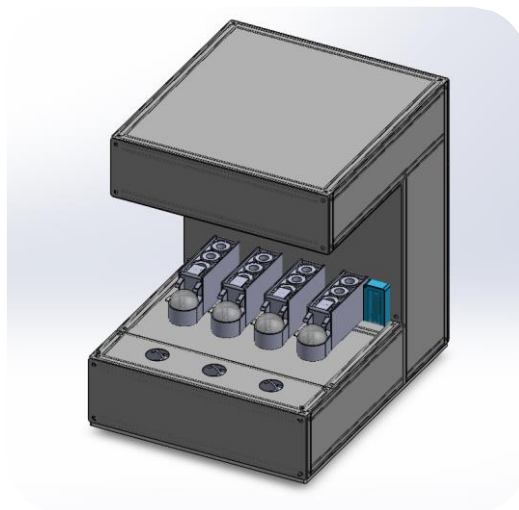


Figure 6: CAD image of exterior sheet metal and 8020 framing over main components

Summary

The Endless Garden's design solution represents a comprehensive integration of mechanical ingenuity and electrical control systems that evolved through systematic iteration to create a truly automated hydroponic platform accessible to users regardless of technical expertise. The plug refill mechanism transformed from complex conveyor and compression spring concepts into a simplified spring-loaded rail system inspired by grocery store drink pushers, with friction calculations informing the critical reduction from 30 to 12 degree angles that enabled smooth plug descent, while reflective IR sensors detect harvest events and Hall effect

sensors monitor supply levels to trigger servo-controlled Pez-dispenser doors that deploy reserve plugs automatically. This automated replacement capability interfaces seamlessly with the nutrient delivery subsystem, where bang-bang controller logic drives relay-controlled peristaltic pumps that maintain optimal 5.8 to 6.0 pH by dispensing acidic, basic, and nutrient solutions from chemically compatible glass jars through silicone tubing into the growing reservoir, with four-phase testing protocols validating individual components before progressing through pH correction loops and full system integration. The dual-reservoir water management system completes the environmental control triumvirate by employing normally closed solenoid valves responding to float sensor signals, allowing gravity-fed flow from backup tanks through check valves that prevent backflow while maintaining the critical 6-inch water depth necessary for root submersion and oxygen exposure, with the normally closed configuration providing inherent safety against flooding during power failures. Plant monitoring controls evolved from sophisticated camera-based tracking and machine learning concepts toward the practical Tredix IR break beam sensor solution, which employs double-check logic to detect when lettuce reaches 6-inch maturity and automatically notifies users without requiring visual inspection or prior growing knowledge, successfully integrating with the Arduino Giga R1 WiFi controller that coordinates all subsystems through shared power rails and centralized monitoring. These interconnected systems mount to a modular 8020 aluminum extrusion frame with sheet metal and acrylic exterior panels, creating a compact platform supporting four simultaneous growing stations with twelve reserve plugs,

Design Process and Results

Introduction

This section provides insight into the iterative development and testing of the Endless Garden's four primary subsystems: plug refill, nutrient controls, water level management, and plant monitoring. The design process for each subsystem required careful consideration of both mechanical integration and electrical control, balancing automation capabilities with system reliability and user accessibility. Through systematic prototyping, component testing, and design refinement, the team addressed challenges ranging from plug stability and sensor selection to pH control precision and water flow management. This resulted in functional subsystems that work together to create a cohesive automated growing environment. The results demonstrate successful mechanical designs, verified electrical integration with the Arduino Giga R1 WiFi controller, and validated sensor performance under simulated operating conditions, positioning the system for full integration and comprehensive testing.

Plug Refill - Mechanical

We went through many iterations of the plug refill design. We mainly focused on how we were going to replace the pods; some ideas included a conveyor belt that moves the pods to place, a compression spring, as well as other concepts having their own challenges. We eventually saw the grocery store bottle mechanism, which we simplified the design and the mechanics, allowing for quick prototyping. We have included our first design iteration, shown in Figure 7.

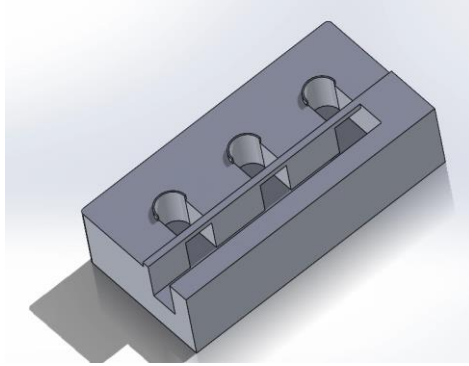


Figure 7: Original Plug Refill Design

The first issue we overcame was that the plugs tipped over easily due to their lightweight nature. To fix this, we designed a guided rail so that the lip of the plugs could slide back and forth easily and eventually slide into the growth pod area. At first the rails were too steep, causing the plugs to not fall into the growing spot. We calculated a more ideal angle by determining the friction force between the plug and the 3D print, leading us to lessen the angle from 30 degrees to around 12 degrees. This change allowed the plugs to fall more smoothly and controlled into the correct spot.

At first a spring-loaded design was going to be used to push the pods forward but since the pods are so light only weighing about .01lbs each the springs tried caused the pod to tip over from the excess force. This led us to go with a gravity fed design where the pods slide along a 12-degree ramp until stopped by a servo motor-controlled door which will be discussed more in the electrical section. The gravity fed design can be shown in Figure 8.

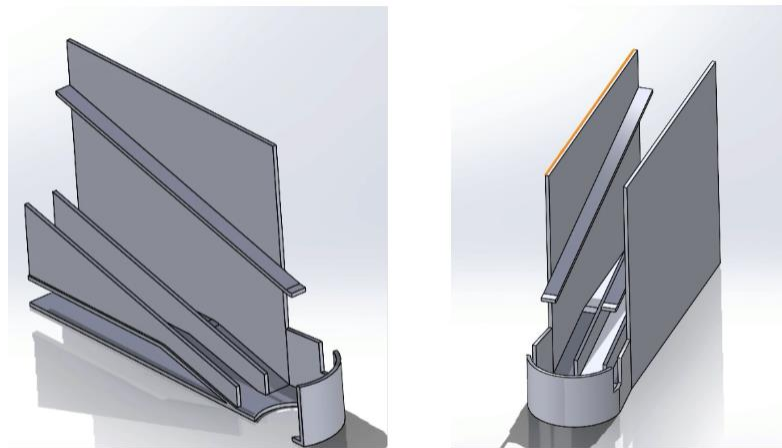


Figure 8: Gravity Fed Plug Refill

The next issue we faced was the ease of plant removal. We also needed to find a way to support the weight of the growing plant, so it does not tip over. At first, we wanted to design the system so the plug would “lock” into place with the prongs. However, we found that these prongs would block the pods path of falling. This led to a new design which would allow the pods to be both supported and allow for ease of removal. Slits will allow the plug remover to sit lower and be held in the correct place. There is a bottom circle which supports the bottom of the pod when it falls into place.

Plug Refill - Electrical

At the beginning of the design process, the team considered four sensor options to detect when a pod had been removed from the system. The goal was to choose a sensor capable of consistently detecting the presence of the pod without requiring significant modifications to the 3D-printed plug refill. The four sensors that were initially considered were: a color sensor, a micro switch, a Hall Effect sensor, and an IR sensor. The color sensor would have detected the pod by identifying its white top edge. However, after consulting with Kevin McCue, we determined that this option was unnecessarily complex for its application and it was eliminated as an option. The micro switch would have been activated by the weight of the pod, but the pod was too light to activate the sensor and was also eliminated as an option. That left the team with the IR and Hall Effect sensors as the best options.

Hall Effect (Uxcell A3144) sensors can detect a magnetic field through plastic, water, and under grow lights with little interference. However, this sensor requires a neodymium magnet to be attached to each pod, which would also be submerged in water. Due to concerns of the neodymium magnet being near the water tank, since neodymium is toxic, and concerns of the magnet interfering with the pods' ability to fall into place, the hall effect sensor was eliminated as an option. However, the hall effect sensor was repurposed to detect if the stations need to be refilled.

The reflective IR sensor (Everlight ITR20001/T) does not require any modifications to the pod, in contrast to how the Hall Effect sensor requires a magnet to be attached to each pod. However, we were worried that the grow lights being above the sensor would interfere with the phototransistor's performance, so we tested the performance of the IR sensor under the grow lights with an environment mimicking the final system conditions, and the sensors functioned normally. Thus, the IR sensors will be used in the final design.

The team decided to use servo motors for the refill and germination system because they have sufficiently precise control and angular movement, because they are straightforward to integrate with an MCU, and because our project only requires the motor to rotate from 0 to 90 degrees. One concern that was raised was that the servo motors might not be strong enough to hold back the weight of all the pods and pusher, but after testing we found that when the servo motors are not powered or connected to ground, they can resist up to 0.4 lbs. before they begin to rotate. The total weight of all four pods and the pusher is 0.07 lbs., which is safely below the 0.4 lbs. that the motor can handle. When the servo motors are powered, they have a stall torque of 1.22 lbf-in, so assuming the motor has an arm length of about 0.34 inches, the motor can resist up to 3.57 lbf.

All the components for this system have been tested, so next steps include mounting the sensors to the plug refill station, completing the wiring of the protoboard, fabricating a housing for the protoboard, and conducting tests of the full completed system.

Nutrient Controls - Mechanical

The fixtures were designed in CAD then tested as shown in Figure 9.

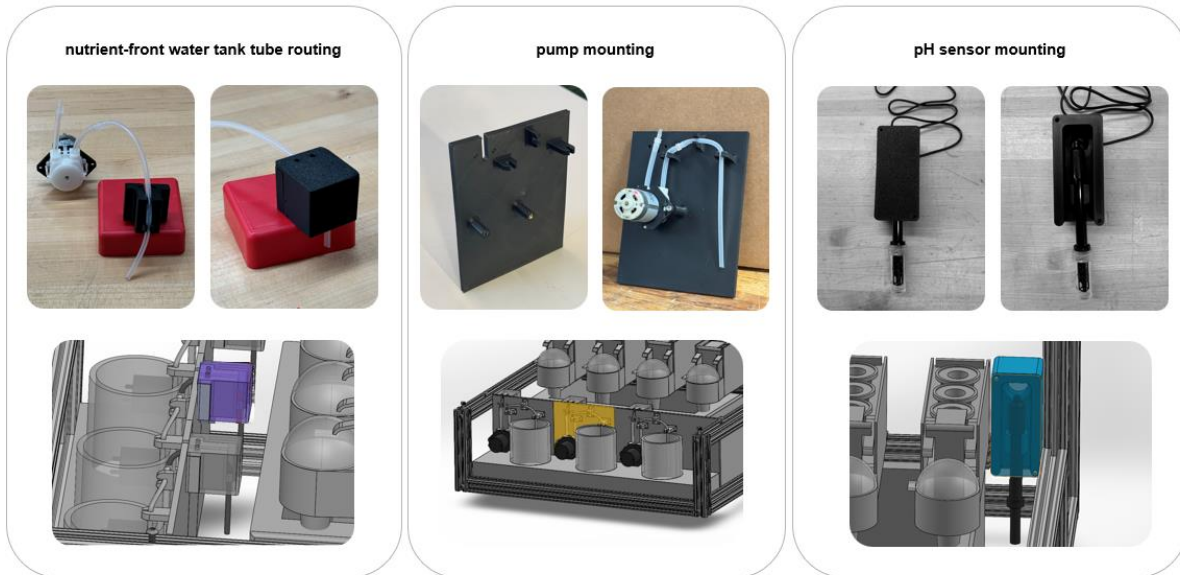


Figure 9: First prototypes of the tube mounting fixture to bring nutrients from the containers to main water tank (left), pump mounting board (middle), and pH sensor fixture (right)

All CAD for current components completed and passed preliminary testing. Additional CAD will be designed for new components once they are selected. Full mounting boards will be 3D printed and assembled by 11/14.

Nutrient Controls - Electrical

After determining the requirements of our system, our team chose to implement a bang-bang controller due to its precision being sufficient for nutrient release into the front tank. This decision led to the development of the circuit diagram for relay logic shown in Figure 10, which maps out the electrical connections needed to control the pumps based on sensor feedback. To ensure proper implementation, the wiring diagram in Figure 11 was created to detail the specific wiring paths between components, allowing the team to systematically connect the Arduino, relays, pumps, and sensors. With these diagrams finalized, the necessary parts were then ordered so testing and implementation could begin.

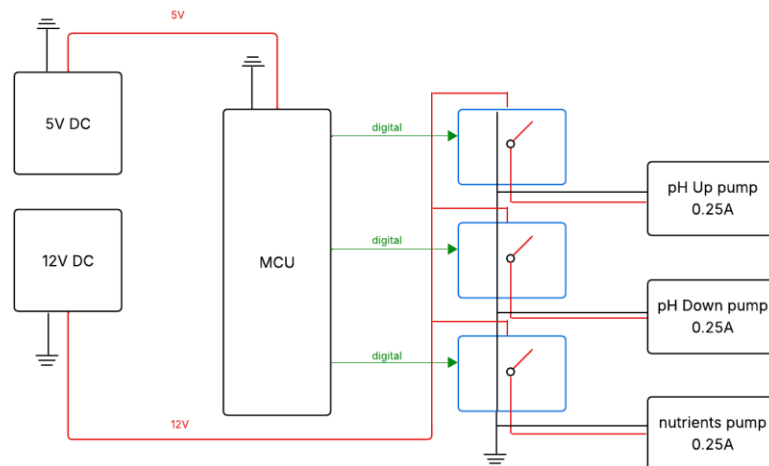


Figure 10: Relay Schematic

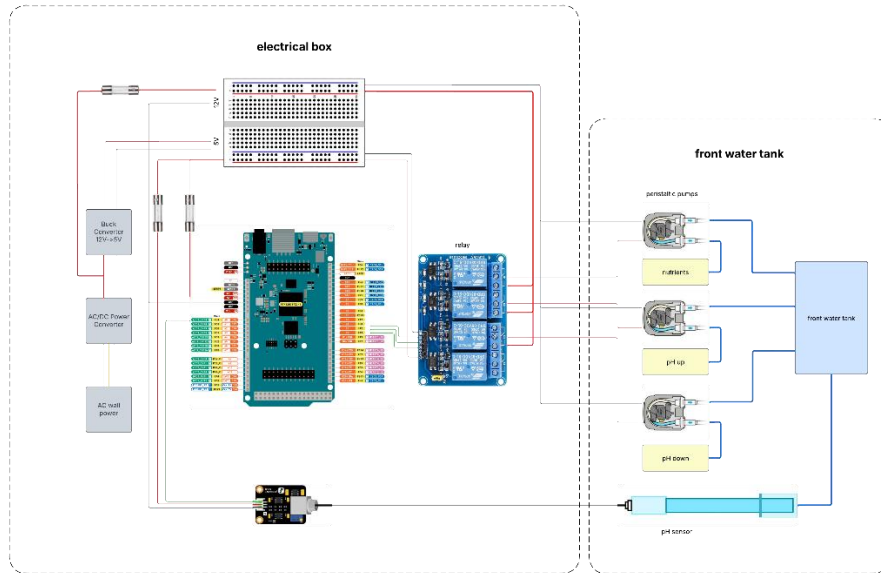


Figure 11: Nutrient Controls Wiring Diagram

The nutrient testing plan followed a four-phase approach that progressively built system complexity as shown in Figure 12. Phase 1 focused on individual component testing, where each pump and sensor was tested separately to verify basic functionality. This included testing the volume flow of the pH up pump, pH down pump, nutrient pump, and pseudo-water tank pump individually, along with validating the pH sensor's accuracy. The pseudo water tank pump worked in place of the real water tank pump that was still being delivered. Phase 2 introduced pH integration by combining the pH sensor with the pH adjustment pumps to test the complete pH control loop. This phase tested both pH correction scenarios: raising pH from 5.9 using the basic solution pump and lowering pH from 5.9 using the acid solution pump. Phase 3 added nutrient integration by testing the nutrient pump in combination with the pseudo-water tank pump to verify they could work together for accurate nutrient dosing. Finally, Phase 4 achieved full integration by combining all previous systems- merging the pH control logic from Phase 2 with the nutrient delivery system from Phase 3, so that the pseudo-water tank pump, pH sensor, pH adjustment pumps, and nutrient pump.

Nutrient Test Plan

Using food coloring to visually see pH up, pH down, and nutrients releasing

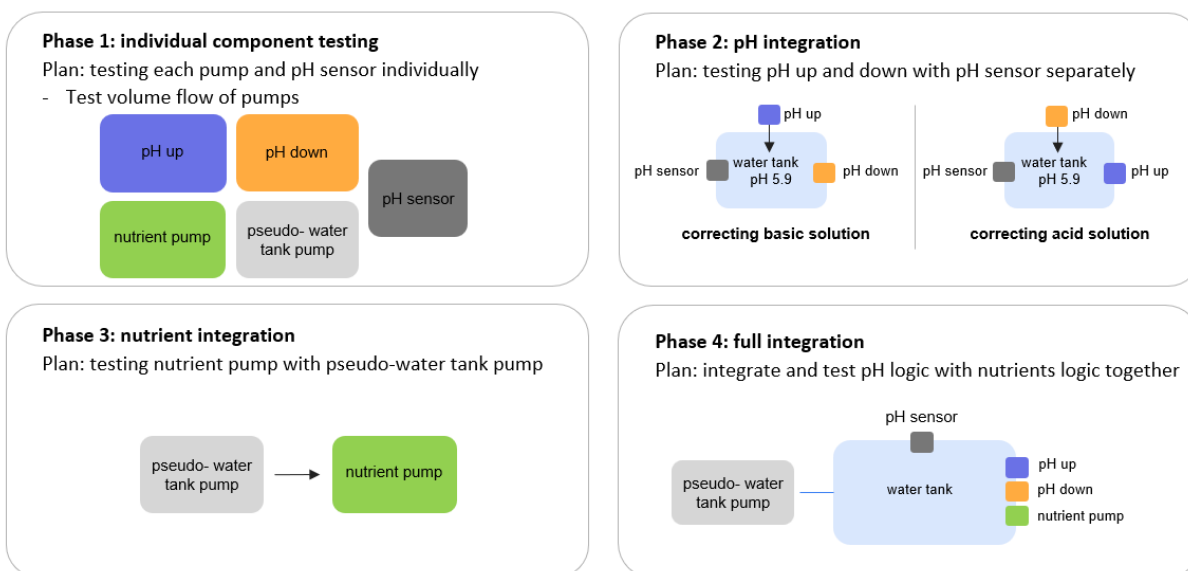


Figure 12: Nutrient Test Plan

All main components for the system have been ordered. Pumps have been operated and tested individually, but the pH sensor still needs to be tested with full system integration. A flow sensor, or a sensor of similar function, as well as additional pumps will be ordered for more reliability and increased precision

Water Level Controls – Electrical

The solenoid valve is powered and controlled through a relay connected to the Arduino. Early in the design phase, our team looked at a few different ways to control the valve, including a direct MOSFET driver and a mechanical float switch. After some research and discussion, we decided that a relay was the best option since it's simple, safe, and reliable for switching a 12V load while keeping the Arduino's logic circuitry isolated. This choice also fits our goal of building a modular system that could tie in easily with the rest of the electronics. Once the relay setup is finalized, we plan to test it on a breadboard to make sure the valve opens and closes properly when the sensor sends its signals. The relay receives a digital "low" or "high" input from the water level sensor. When the signal is low, the Arduino powers the relay, which sends 12V to open the solenoid valve. When the sensor detects that the water is full, the Arduino cuts power, closing the valve. This simple on-off setup gives us a dependable control method, similar to how the nutrient pH system operates. The wiring also connects to the same power rail and microcontroller used by the nutrient and plug refill subsystems, keeping everything neat and consistent.

When developing this system, we focused on finding the right balance between accuracy, reliability, and ease of building. We brainstormed a few different ideas, like using a pump that ran on a timer or a mechanical float valve that would open automatically. Both had downsides. The float valve did not offer enough precision, and the timed pump design added unnecessary complexity to the code. Since hydroponic systems are sensitive to small water-level changes, we wanted something that could react quickly and predictably without a lot of tuning. The electronic solenoid setup ended up being the most practical choice; it is simple, accurate, and it lets us track refill activity through the Arduino for future improvements.

The final water level control design gives us a reliable and automated way to keep the growing reservoir at the right height. Using a gravity-fed top tank, a check valve, and an electronically controlled solenoid, the

system keeps the bottom reservoir full without needing the user to manually top it off. Maintaining a consistent 6-inch water level ensures the roots stay submerged while still getting enough oxygen, which helps the plants grow healthier and stronger over time. Overall, the design decisions came down to creating something that's easy to maintain, accurate for hydroponic use, and dependable enough to run continuously.

All CAD has been completed, and reservoirs have been ordered. An electronic flow subsystem between reservoirs has also been laid out and is awaiting approval from Kevin. Once Kevin verifies the electrical components, they will be ordered, and testing will follow shortly after that.

Plant Monitoring Controls – Electrical

Multiple options were looked at when it came to the monitoring of the plant. Some options included using an integrated camera that would record the plant's growth and create a logarithmic curve of its growth cycle. Another option was to integrate machine learning and computer vision into the design that would allow the system to inform the user when the ideal moment was to stop the growth. Due to the limitations of Arduino's on-board memory, it was decided that a simpler, practical solution would be more ideal to implement. A break beam sensor was found to fulfil the requirements of the design. This would allow seamless integration with the design and would also allow future changes if the plants were to change by allowing different height options.

The ideal choice based on the design was the Tredix IR break beam sensor, as it fits the most important markers with an effective range of 1m, a length greater than the footprint of the device. The logic was designed to help prevent any false readings from occurring as seen in Figure 13 below.

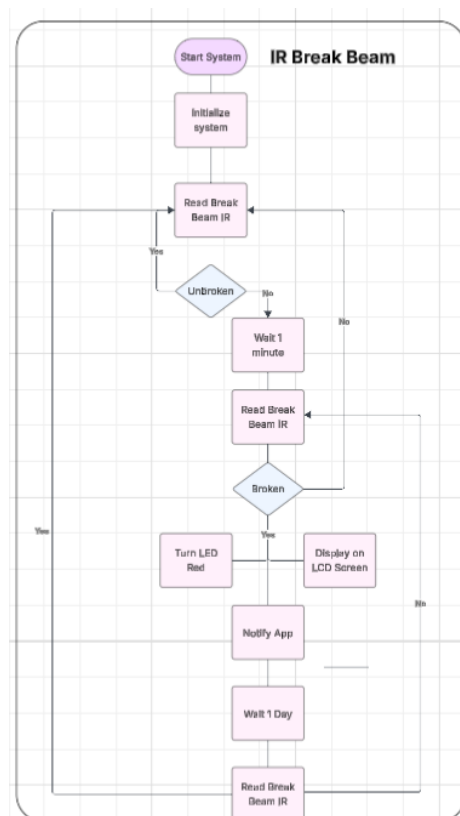


Figure 13: Logic of IR break beam sensor

The introduction of a double check mechanism was implemented to help ensure that the user did not accidentally set the system off, as the growth of the plants was so slow that having the right allowance for minute movements helped decrease the need for user interaction.

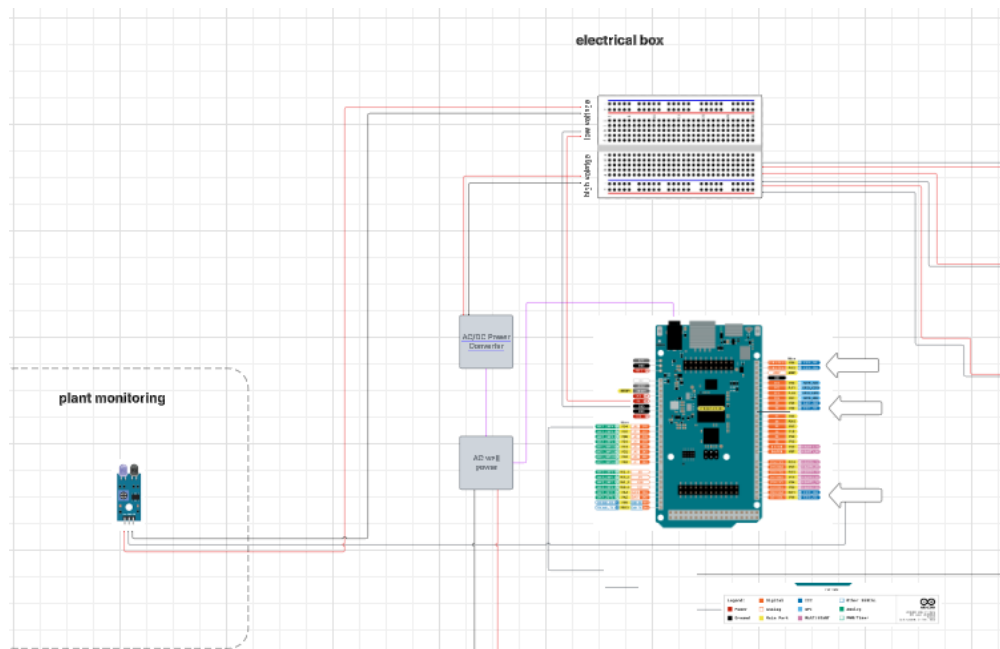


Figure 14: Wiring Diagram

The output of the receiver gets sent directly in a single digital pin through the Arduino Giga R1 as seen in Figure 14 above.

The IR break beam sensor was integrated into the Arduino Giga R1 successfully; a code was devised using the respective logic. The IR break beam sensor was tested at an effective distance used on the system by measuring 16 inches apart. The system responded with a message that was programmed informing that the break beam sensor had broken. The next steps are to integrate it in with the 8020 frame and the plug refill system.

Summary

The design and testing process for the Endless Garden's subsystems demonstrates systematic problem-solving and iterative refinement that transformed conceptual challenges into functional solutions. The plug refill mechanism evolved from complex conveyor and compression spring concepts to a simplified grocery store bottle pusher design, overcoming plug tipping issues through guided rail implementation and angle optimization from 30 to 12 degrees based on friction calculations, while sensor selection narrowed from four options to the reflective IR sensor after eliminating color sensors, micro switches, and Hall effect sensors due to complexity, insufficient sensitivity, and water toxicity concerns respectively. Nutrient controls progressed through bang-bang controller implementation with relay logic schematics and comprehensive wiring diagrams, followed by a four-phase testing plan that systematically validates individual pump functionality, pH control loops, nutrient dosing accuracy, and full system integration. Water level management balanced accuracy, reliability, and simplicity by selecting relay-controlled solenoid valves over MOSFET drivers and mechanical float switches, creating a gravity-fed dual-reservoir system that maintains consistent 6-inch water depth through Arduino-monitored on-off control. Plant monitoring advanced from complex camera-based logarithmic tracking and machine learning concepts to the practical Tredix IR break beam sensor with double-check logic that prevents false readings while

accommodating slow plant growth, successfully integrating with the Arduino Giga R1 and validating 16-inch detection distance under simulated operating conditions.

Results

The Endless Garden prototype has achieved significant progress across all subsystems, with several components reaching functional testing stages and others advancing toward full integration. The plug refill mechanism successfully demonstrates sequential pod replacement through its rail and wall design, allowing reserve plugs to fall into position one at a time while supporting mature plants during growth; three reserve plugs remain staged behind a servo-controlled door awaiting automated deployment, though IR and Hall effect sensor testing is pending component arrival to finalize pod detection capabilities. Nutrient control subsystems have completed CAD design and passed preliminary fixture testing, with mounting boards scheduled for assembly; individual pump functionality has been verified, though pH sensor integration and additional flow monitoring components remain in development to enhance system precision and reliability. Water level controls have finalized mechanical design with reservoirs ordered and an electronic flow subsystem between tanks awaiting advisor approval before component procurement and testing commence. Plant monitoring controls have achieved the most complete integration, with the IR break beam sensor successfully incorporated into the Arduino Giga R1 and validated at the required 16 inch detection distance, accurately triggering programmed harvest notifications; final integration with the 8020 frame and plug refill assembly represents the remaining implementation step for this subsystem.

Summary and Impact

Working on Endless Garden has been a rewarding experience that has pushed our team to think creatively about how engineering can solve everyday problems. We set out to design an indoor gardening system that makes growing fresh produce accessible to anyone, regardless of time, space, or experience. Along the way, we learned how to merge mechanical design, electrical control, and software logic into one reliable and easy-to-use system. Seeing our design evolve from sketches to a functioning prototype reminded us that even small ideas can grow into something meaningful when collaboration and persistence are at the core.

Our system addresses a real gap in current indoor gardening products by providing a continuous harvest instead of a batch cycle. This approach gives users a consistent source of fresh produce without long waiting periods or complicated maintenance. By automating key functions such as plug replacement, nutrient regulation, and water level control, we were able to reduce user input while maintaining plant health. The result is a compact and sustainable system that demonstrates how thoughtful engineering can make personal food production simpler and more dependable.

This project also taught us valuable lessons about problem solving, prototyping, and teamwork. We learned how to test and refine individual subsystems, troubleshoot unexpected challenges, and apply feedback from advisors to make our design stronger. These skills will carry over into our professional careers, where solving complex problems with creativity and practicality is essential.

Looking ahead, the Endless Garden could serve as a foundation for future research or product development. Adding wireless monitoring, nutrient optimization, or machine learning could transform it into a fully intelligent gardening platform. Beyond personal use, it could be adapted for schools, small restaurants, or research labs that want to grow food efficiently in limited space.

In the end, the Endless Garden represents more than just a class project; it's a small step toward a future where sustainable food production is part of everyday life. Our team hopes that what we've built inspires others to think differently about how engineering can make sustainable living more achievable, one small system at a time.

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Appendix: Engineering Analysis

Appendix A: Servo motor force calculations

As stated in the “design process and results: plug refill electrical” section, some concerns were raised about the motor's ability to hold its position with some weight pushing on it. The total weight pushing on the motor is $W_{total} = 0.073237592 \text{ lbs}$. The stall torque of the servo motor (SG90 Micro Servo Motor by Miuzei) when it's powered at 4.8 V is $\tau = 1.40 \text{ kgcm}$. The four radiuses represent different places where the motor could be attached to the door using galvanized steel wire. These calculations show that regardless of the steel wires' distance from the axis of rotation, the motor will be able to hold its position with the weight of the system pushing on it.

When the motor is powered and connected to ground:

$$\tau = 1.40 [\text{kgcm}] \cdot \frac{9.807 \text{ N}}{1 \text{ kg}} \cdot \frac{1 \text{ m}}{100 \text{ cm}} = 0.137298 \text{ Nm}$$

$$r_1 = 0.14 \text{ in} \cdot \frac{1 \text{ m}}{39.37 \text{ in}} = 0.003556 \text{ m}$$

$$r_2 = 0.24 \text{ in} \cdot \frac{1 \text{ m}}{39.37 \text{ in}} = 0.006096 \text{ m}$$

$$r_3 = 0.34 \text{ in} \cdot \frac{1 \text{ m}}{39.37 \text{ in}} = 0.008636 \text{ m}$$

$$r_4 = 0.44 \text{ in} \cdot \frac{1 \text{ m}}{39.37 \text{ in}} = 0.011176 \text{ m}$$

$$F_1 = \frac{\tau}{r_1} = \frac{0.137298 \text{ Nm}}{0.003556 \text{ m}} = 38.6102 \text{ N} \cdot \frac{1 \text{ lbf}}{4.448 \text{ N}} = 8.68034 \text{ lbf}$$

$$F_2 = \frac{\tau}{r_2} = \frac{0.137298 \text{ Nm}}{0.006096 \text{ m}} = 22.5226 \text{ N} \cdot \frac{1 \text{ lbf}}{4.448 \text{ N}} = 5.06353 \text{ lbf}$$

$$F_3 = \frac{\tau}{r_3} = \frac{0.137298 \text{ Nm}}{0.008636 \text{ m}} = 15.8983 \text{ N} \cdot \frac{1 \text{ lbf}}{4.448 \text{ N}} = 3.57426 \text{ lbf}$$

$$F_4 = \frac{\tau}{r_4} = \frac{0.137298 \text{ Nm}}{0.011176 \text{ m}} = 12.2851 \text{ N} \cdot \frac{1 \text{ lbf}}{4.448 \text{ N}} = 2.76193 \text{ lbf}$$

$$W_{pods} = 4 \cdot 6.63 \text{ g} \cdot \frac{1 \text{ lb}}{453.6 \text{ g}} = 0.058466592 \text{ lbs}$$

$$W_{pusher} = 6.70 \text{ g} \cdot \frac{1 \text{ lb}}{453.6 \text{ g}} = 0.014771 \text{ lbs}$$

$$W_{total} = W_{pods} + W_{pusher} = 0.058466592 \text{ lbs} + 0.014771 \text{ lbs} = 0.073237592 \text{ lbs}$$

Figure 15: Servo motor maximum force calculations

Appendix B: Nutrient Controls Logic Calculations

Calculations were done for the nutrient control subsystem to ensure that the dispensing rate on nutrients has no larger than a 1% error. Through these calculations we confirmed that the current system falls within the requirements of the project.

Error calculations for nutrient dispensing

Front tank dimensions: 12"W × 6"H × 18"L

Total tank volume:

- $12 \times 6 \times 18 = 1,296$ cubic inches
- $1,296 \text{ in}^3 \div 231 \text{ in}^3/\text{gal} = 5.610390 \text{ gal}$ or 21.237662 L

Water level 2 inches from top:

- Working height: $6" - 2" = 4"$
- Working volume: $12 \times 4 \times 18 = 864 \text{ in}^3$
- $864 \text{ in}^3 \div 231 = 3.740260 \text{ gal}$ or 14.158442 L

Accounting for pump and air stone (~0.5L displaced):

- Actual water volume: $14.158442 - 0.5 = 13.658442 \text{ L}$

Target nutrient concentration:

- 2.5 ml nutrients per 4 L water
- For 13.658442 L: $(13.658442 \div 4) \times 2.5 = 8.536526250 \text{ ml}$ total nutrients needed

Dosing with current system:

- Every 250 mL water added = 0.15625 mL nutrients
- Over 13.658442 L: $(13.658442 \text{ L} \div 0.25 \text{ L}) \times 0.15625 \text{ mL} = 8.536526563 \text{ mL}$ nutrients delivered

Error calculation:

- Target: 8.536526250 mL
- Delivered: 8.536526563 mL
- Percent Error: $3.7 \times 10^{-6} \%$

Figure 16: Error calculations for nutrient dispensing

Appendix C: Plug Refill

The initial 30° rail angle resulted in pods sliding too quickly and inconsistently into the growing position. Force analysis showed that at 30°, the pod experienced an acceleration of 2.78 m/s² and reached an impact velocity of 1.04 m/s, causing the pod to not land correctly in the growing position. We calculated the forces acting on the pod: weight $W = mg = 0.441$ N, normal force $N = W \cdot \cos(\theta)$, friction force $f = \mu \cdot N$, and parallel force $F_p = W \cdot \sin(\theta)$. For the 30° angle, $F_p = 0.221$ N exceeded static friction $f_s = 0.153$ N, allowing the pod to slide on gravity alone. The 12° angle was able to gently guide the pods in the place. Testing confirmed the 12° angle worked and allowed the pods to correctly fall into place.

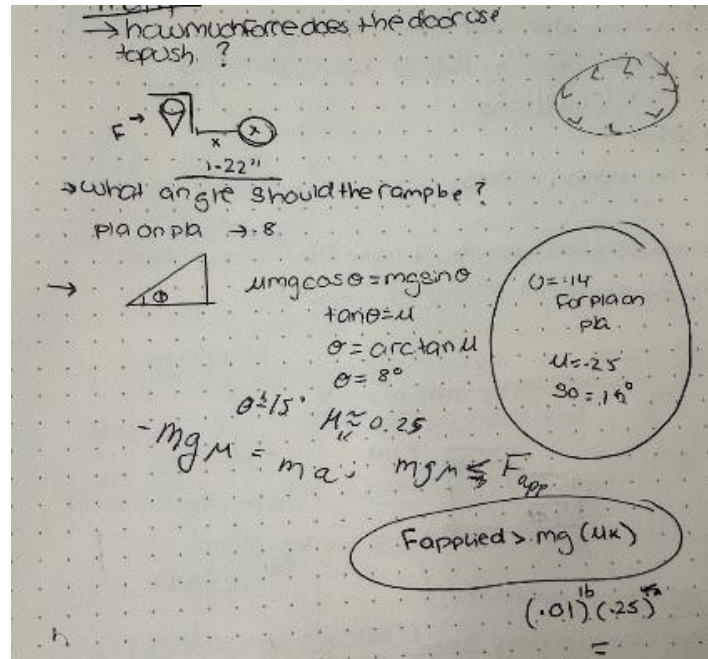


Figure 17: Plug Refill ramp Angle calculations

Appendix D: Reflective IR sensor and servo motor schematic

Figure 18 shows a basic schematic for how the reflective infrared (IR) sensors (Everlight ITR20001/T), SG90 servo motors, and NPN transistors are connected to perform the pod-refill sequence. We created this schematic to understand how our system would work before we started building.

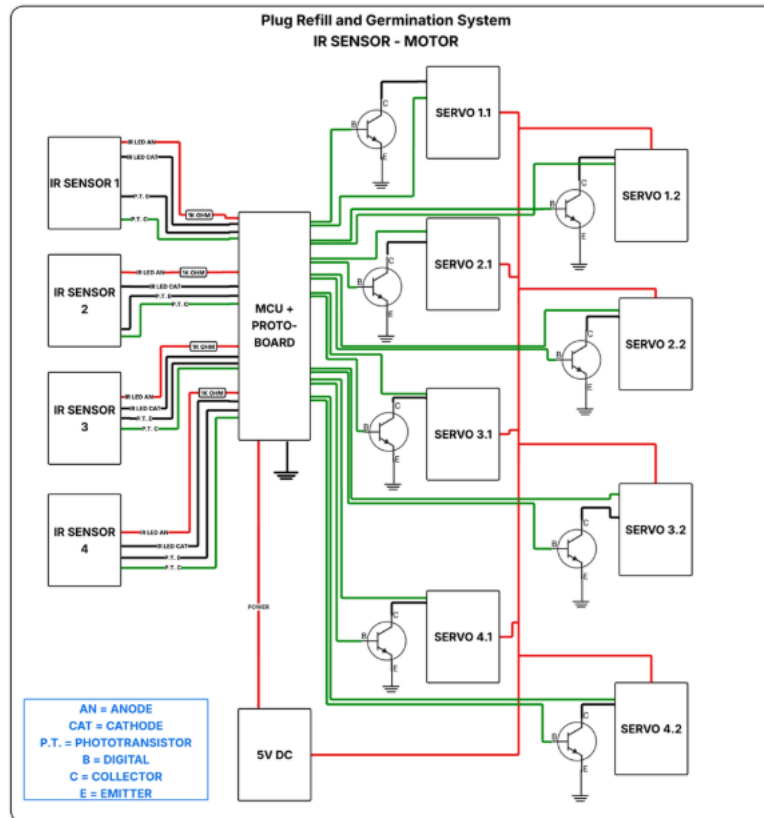


Figure 18: IR sensor, servo motor, and transistor schematic

Appendix E: Wiring diagram showing motor and NPN transistor connections

Figure 19 shows a wiring diagram of the servo motors and NPN transistors connected to a protoboard and MCU. We created this wiring diagram to understand how each component would be wired and soldered before we started the processes of building and testing.

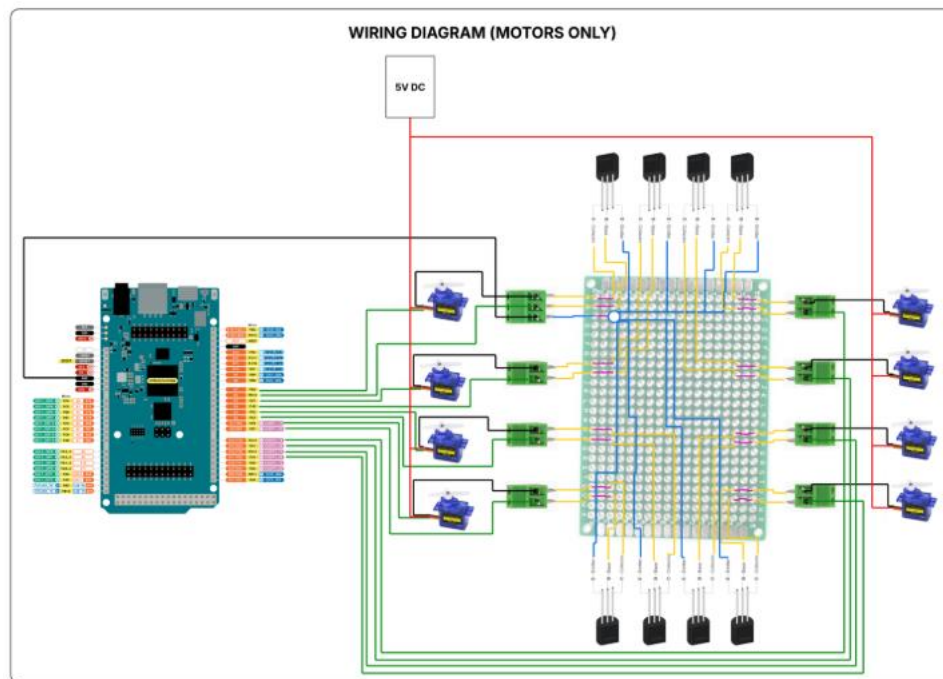


Figure 19: Servo motor and transistor wiring diagram